Project Management

7. Cost Management

Contents

- What is Project Cost Management?
  - Types of Cost
- Estimate Cost
- Determine Budget
- Control Cost
  - What is Earned Value?
  - EV related calculation.

Cost Management

Every project boils down to money.

What is Project Cost Management

- Project cost is the ***cost of the resources*** needed to complete project activities.
- Cost Management should consider the effect of project decisions on the subsequent recurring cost of using, maintaining, and supporting the product, service, or result of the project.

Types of Cost

Direct costs

- Costs incurred only because we started the project (Wages, Material, Equipment etc).

Indirect costs

- Cost that be passed on to the project by prior agreement, shared by multiple projects (taxes, training, project management software license, and so on).

Variable costs

- Costs that vary depending on the amount of work or production (Cost of materials, supplies, wages etc..).

Fixed costs

- These costs remain constant throughout the project (Cost of office setup, rentals etc...).

Plan Cost Management

Just like all of the other knowledge areas, you need to plan out all of the processes and methodologies you'll use for Cost Management up front.

Estimate Costs

This means figuring out exactly how much you expect each work activity you are doing to cost. So each activity is estimated for its time and materials cost, and any other known factors that can be figured in.

Determine Budget

Here's where all of the estimates are added up and baselined. Once you have figured out the baseline, that's what all future expenditures are compared to.

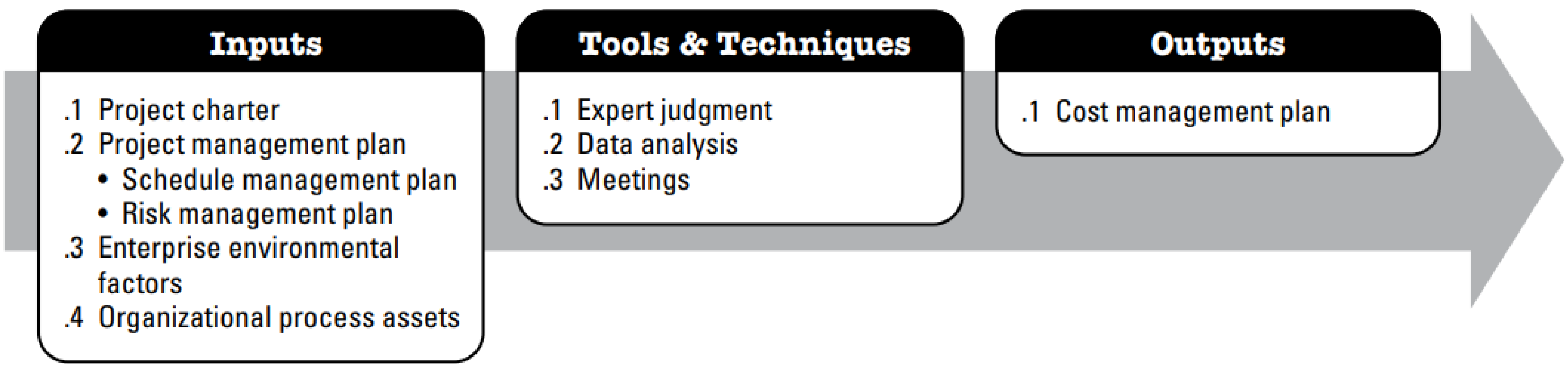
Control Costs

This just means tracking the actual work according to the budget to see if any adjustments need to be made.

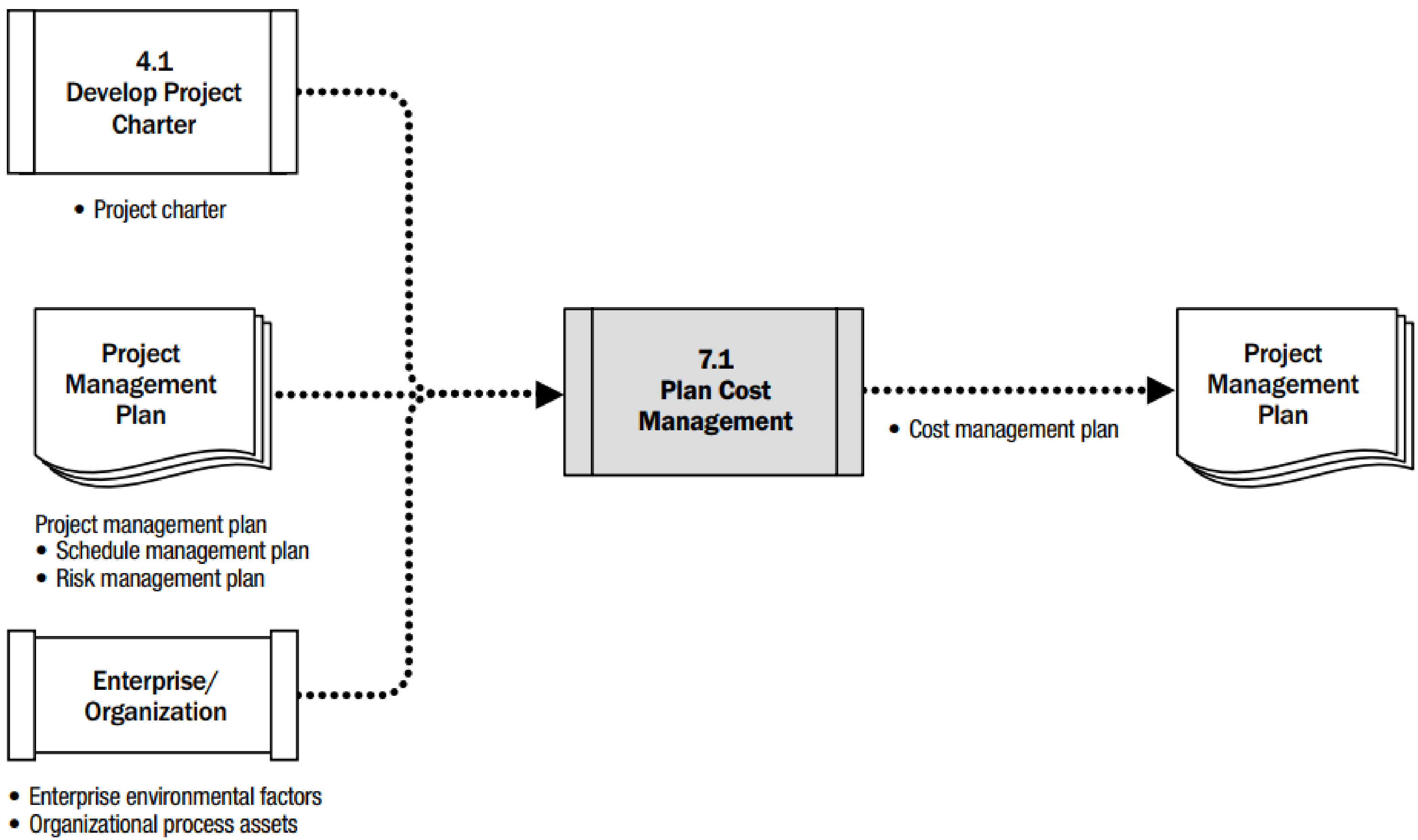
Plan Cost Management

Plan Cost Management

- **What**: process of defining how the project costs will be estimated, budgeted, managed, monitored, and controlled.
- **Why:** guidance and direction on how the project costs will be managed throughout the project.
- **When:** performed once or at predefined points in the project.



Data Flow


4.1
Develop Project
Charter
• Project charter
Project
Management
Plan
Project management plan
• Schedule management plan
• Risk management plan
Enterprise/
Organization
• Enterprise environmental factors
• Organizational process assets
7.1
Plan Cost
Management
• Cost management plan
Project
Management
Plan

Tools & techniques

Expert Judgment

- Any group or person with expertise in developing Cost Management Plans

Meetings

- Participants with responsibility for cost planning and execution: project team, project manager, sponsor, ..

Data Analysis

- Alternatives analysis can include reviewing strategic funding options such as: self-funding, funding with equity, or funding with debt.
- Consideration of ways to acquire project resources such as making, purchasing, renting, or leasing

Main steps of alternatives analysis

- **Identify Alternatives**: List possible methods, technologies, materials, or vendors.
- **Estimate Costs**: Calculate direct and indirect costs for each option.
- **Assess Non-Cost Factors**: Quality, risk, compliance, resource fit, etc.
- **Compare Options**: Use a decision matrix, scoring model, or weighted criteria.
- **Select the Best Fit**: Choose the option offering the best balance between cost and value.

Alternatives analysis

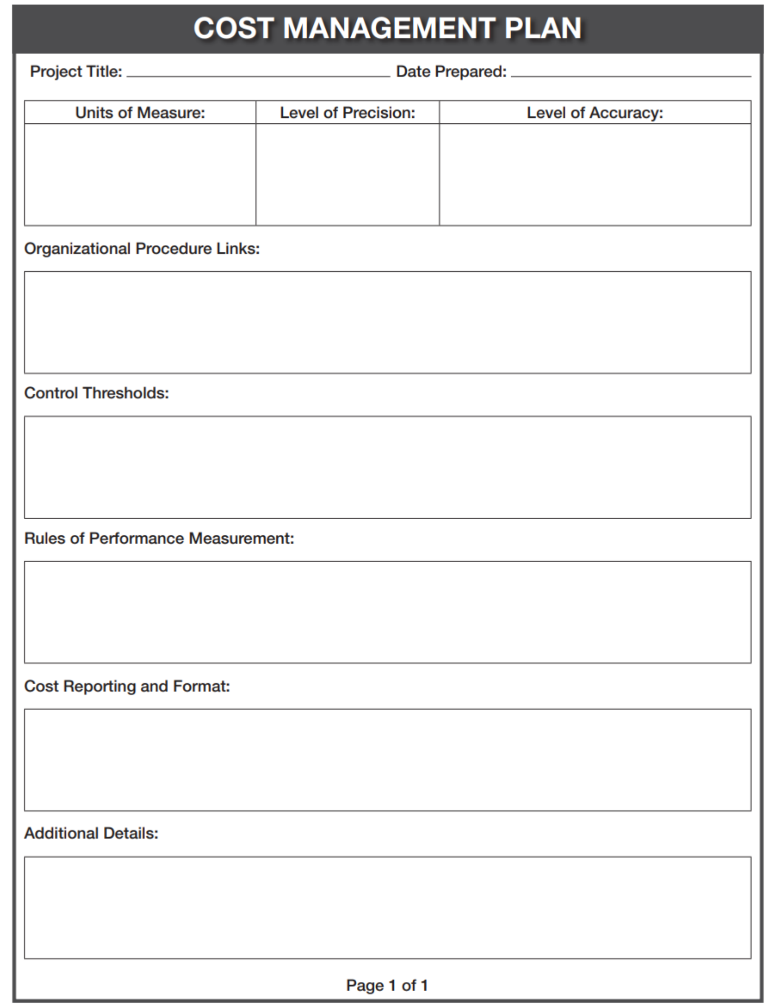
Criteria	Weight (%)	Option A: In-house Dev	Option B: Outsourced	Option C: SDK Tool
Initial Cost ($)	20%	12,000	8,000	5,000
Time to Implement (Weeks)	15%	6	4	2
Risk Level	20%	Low	Medium	Low
Quality/Flexibility	20%	High	Medium	Low
Total Cost of Ownership	15%	15,000	10,000	7,000
Compliance/Security Fit	10%	High	Low	Medium
Weighted Score	100%	TBD	TBD	TBD

Outputs

Cost management plan

Describes how the project costs will be planned, structured, and controlled.

- **Units of measure**. unit used in measurements (such as staff hours, staff days; meters, liters, currency form)
- **Level of precision, accuracy**.
- **Control thresholds** (%). Variance thresholds for monitoring cost performance
- **Rules of performance measurement**. Earned value management (EVM) rules of performance measurement



COST MANAGEMENT PLAN

Project Title: ______ Date Prepared: ______

Units of Measure:	Level of Precision:	Level of Accuracy:

Organizational Procedure Links:

Control Thresholds:

Rules of Performance Measurement:

Cost Reporting and Format:

Additional Details:

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Estimate Cost

Estimate Cost

- **What**: process of developing an ***approximation*** of the cost of resources needed to complete project work.
  - Prediction is based on the information known at a given point in time
  - Cost trade-offs and risks should be considered, such as make versus buy, buy versus lease, and the sharing of resources in order to achieve optimal costs for the project.
- **Why:** determines the monetary resources required for the project.
- **When:** performed periodically throughout the project as needed.
  - Cost estimates should be reviewed and refined during the course of the project to ***reflect additional detail*** as it becomes available and assumptions are tested.
  - The accuracy of a project estimate will increase as the project progresses through the project life cycle.
    - Rough Order of Magnitude (ROM): −25% to +75%
    - Definitive estimates: −5% to +10%

Process

Inputs

.1 Project management plan
- Cost management plan
- Quality management plan
- Scope baseline

.2 Project documents
- Lessons learned register
- Project schedule
- Resources requirements
- Risk register

.3 Enterprise environmental factors

.4 Organizational process assets

Tools & Techniques

.1 Expert judgment

.2 Analogous estimating

.3 Parametric estimating

.4 Bottom-up estimating

.5 Three-point estimating

.6 Data analysis
- Alternatives analysis
- Reserve analysis
- Cost of quality

.7 Project management information system

.8 Decision making
- Voting

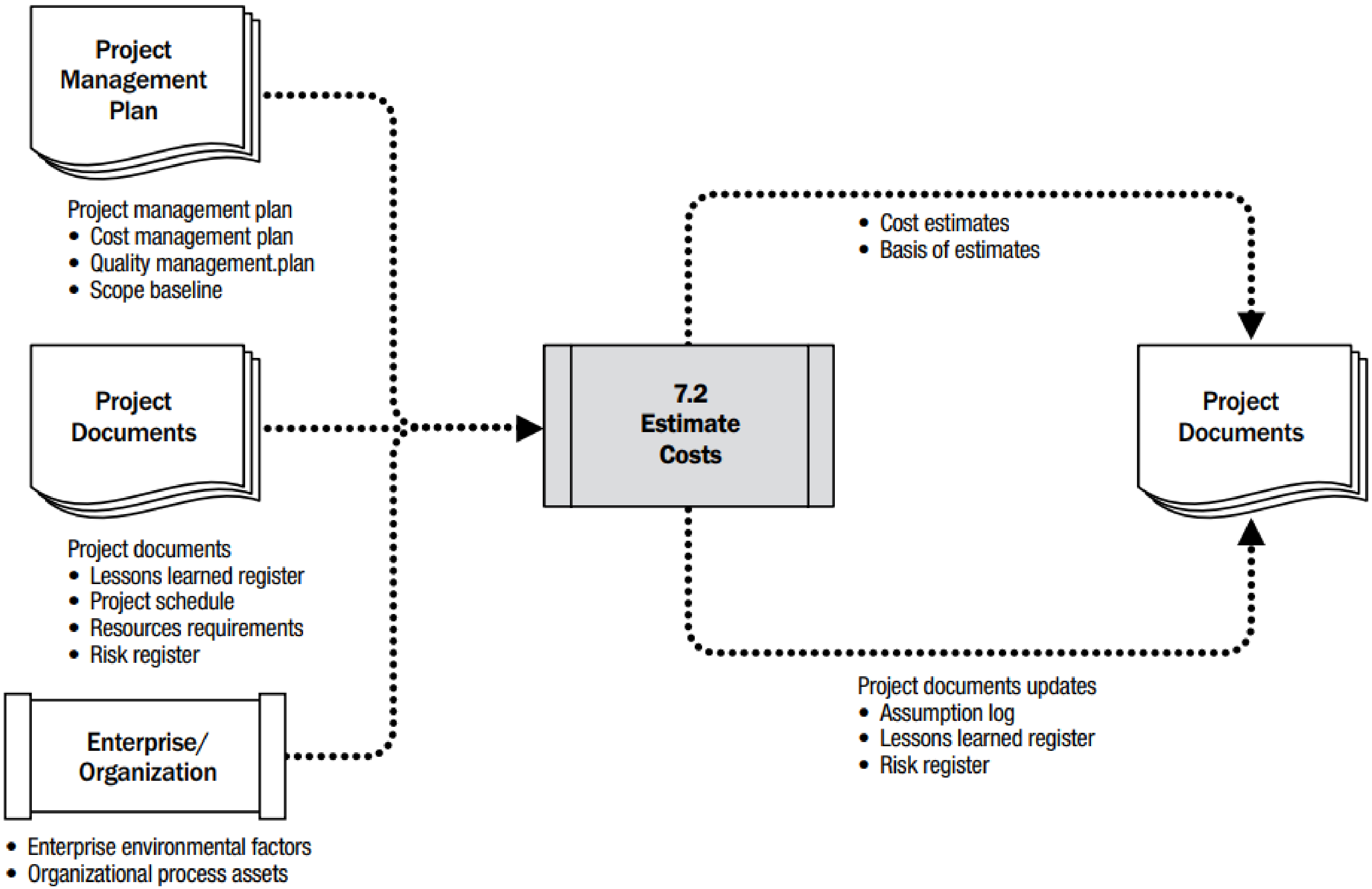
Outputs

.1 Cost estimates

.2 Basis of estimates

.3 Project documents updates
- Assumption log
- Lessons learned register
- Risk register

Data Flow



Project Management Plan
Project management plan
• Cost management plan
• Quality management.plan
• Scope baseline
Project Documents
Project documents
• Lessons learned register
• Project schedule
• Resources requirements
• Risk register
Enterprise/ Organization
• Enterprise environmental factors
• Organizational process assets
7.2 Estimate Costs
• Cost estimates
• Basis of estimates
Project Documents
Project documents updates
• Assumption log
• Lessons learned register
• Risk register

Tamika and Sue are going to open a new café, and their PM Alice need to estimate the project cost.

A. Bottom-up estimating

B. Analogous estimating

C. Expert judgment

D. Parametric estimating

E. Three-point estimates

1. The café across the street opened just a few months ago. Alice sits down with the contractor who did the work there and asks him to help her figure out how much it will cost. He takes a look at the equipment Sue and Tamika want to buy and the specs for the cabinets and seating and tells her what she can afford to do with the budget she has.

Tool: ....

2. Alice creates a spreadsheet with all of the historical information from similar remodeling projects that have happened on her block. She sits down and types in Tamika and Sue's desired furnishings and the square footage of the room to generate an estimated cost.

Tool: ....

3. Before Alice finishes her schedule, she gathers all of the information she has about previous projects' costs (like how much labor and materials cost). She also talks to a contractor, who gives valuable input.

Tool: ....

4. Alice sits down and estimates each and every activity and resource that she is going to need. Then she adds up all of the estimates into "rolled-up" categories. From there she adds up the categories into an overall budget number.

Tool: ....

5. Tamika sets up an appointment with the same contractor her friend used for some remodeling work. The contractor comes to the house, takes a look at the room, and then gives an estimate for the work.

Tool: ....

6. Alice figures out a best-case scenario, a most likely scenario, and a worst-case scenario. Then she uses a formula to come up with an expected cost for the project.

Tool: ....

Tools & Techniques

1: **B. Analogous estimating**

Since Alice is using the contractor's experience with a similar project to figure out how long her project will take, she is assuming that her project will go like the café one did.

2: **D. Parametric estimating**

In this one Alice is just applying some numbers particular to her project to some historical information she has gathered from other projects and generating an estimate from that.

3: **C. Expert judgment**

Expert judgment often involves going back to historical information about past projects as well as consulting with experts or using your own expertise.

4: **A. Bottom-up estimating**

Starting at the lowest level and rolling up estimates is bottom-up estimating. Alice started with the activities on her schedule and rolled them up to categories and finally to a budget number.

5: **C. Expert judgment**

This is another example of asking somebody who has direct experience with this kind of work to give an estimate.

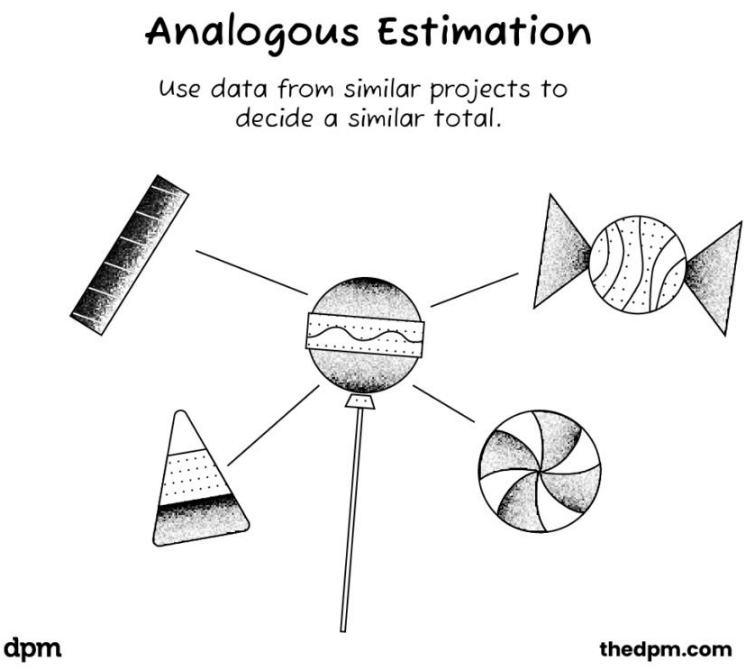
6: **E. Three-point estimate**

Alice came up with the three estimates and then performed a triangular distribution calculation.

Tools & Techniques

Analogous Estimating

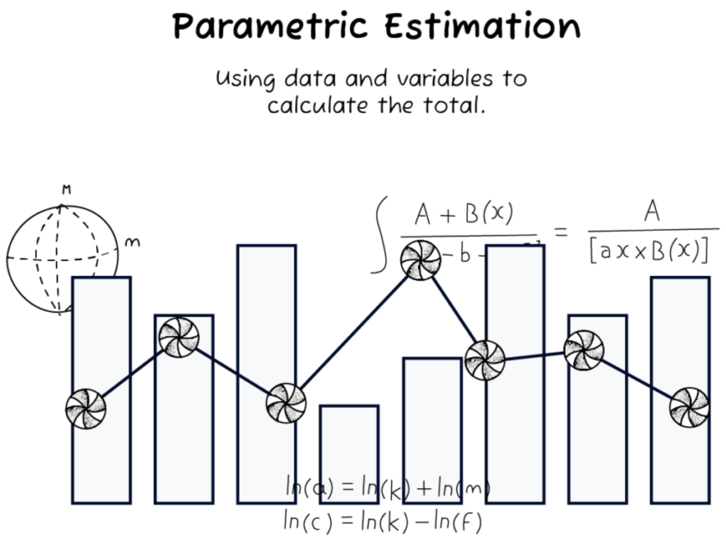
- Uses information from a previous, similar project, such as duration, budget, size and complexity for future project.
- Analogous estimate is generally ***less costly and time consuming but generally less accurate***.
- This estimate will be more accurate if previous project is similar in nature and not just in appearance.
- Analogous duration estimating is frequently used to estimate project duration when ***there is a limited amount of detailed information*** about the project.



Tools & Techniques

Parametric Estimating

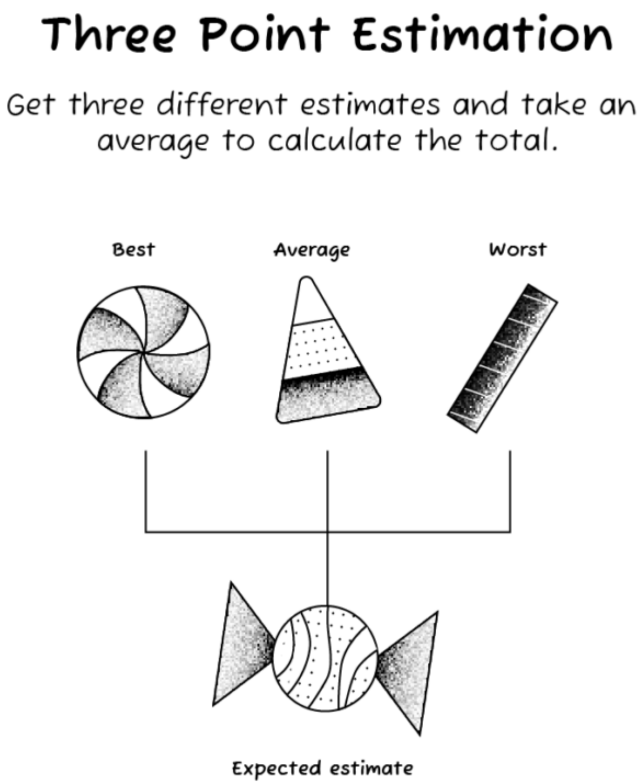
- Parametric estimate uses a statistical relationship between historical data and other variables.
  - Identify a measurable parameter (e.g., cost per unit, hours per line of code, dollars per square meter).
  - Use a formula or rate based on historical data or benchmarks.
  - Multiply the parameter by the quantity needed for the current project.
- Example - A resource will take 20hrs per module and hence 50 modules will take 1000hrs (50X20 = 1000hrs)



Tools & Techniques

Three point estimates (PERT)

- When there is insufficient historical data or when using judgmental data.
- A three-point estimate uses average of optimistic, most likely, and pessimistic estimates and hence improving the accuracy.
    - **Pessimistic (cP)** assumes the worst case scenario
    - **Most likely (cM)** – The realistic and most likely estimate
    - **Optimistic (cO)** is the best case scenario.
- *cE = (cO + cM + cP)/3*



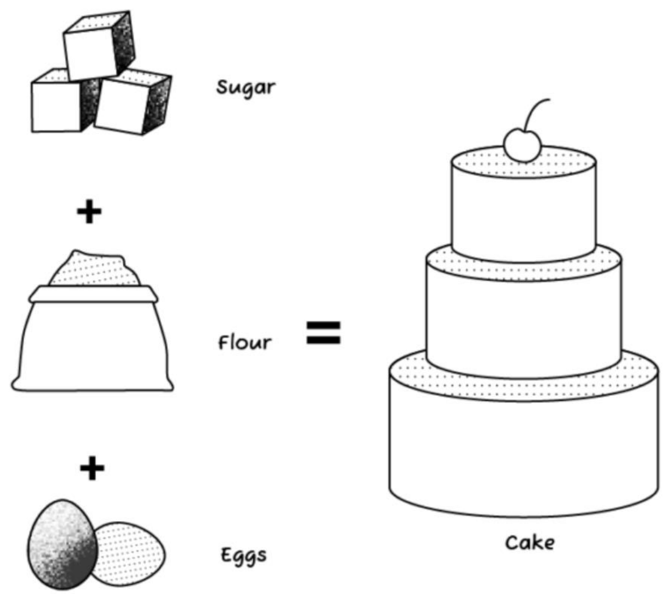
Beta distribution

(Optimistic duration + 4 Most likely duration + Pessimistic duration) ÷ 6 = Expected duration

Tools & Techniques

Bottom-up Estimating

- The work within the activity is decomposed into more detail.
- Once these estimations have been performed, the pieces may be summed up from the bottom back to activity level.
- The most accurate



Tools & Techniques: Data Analysis

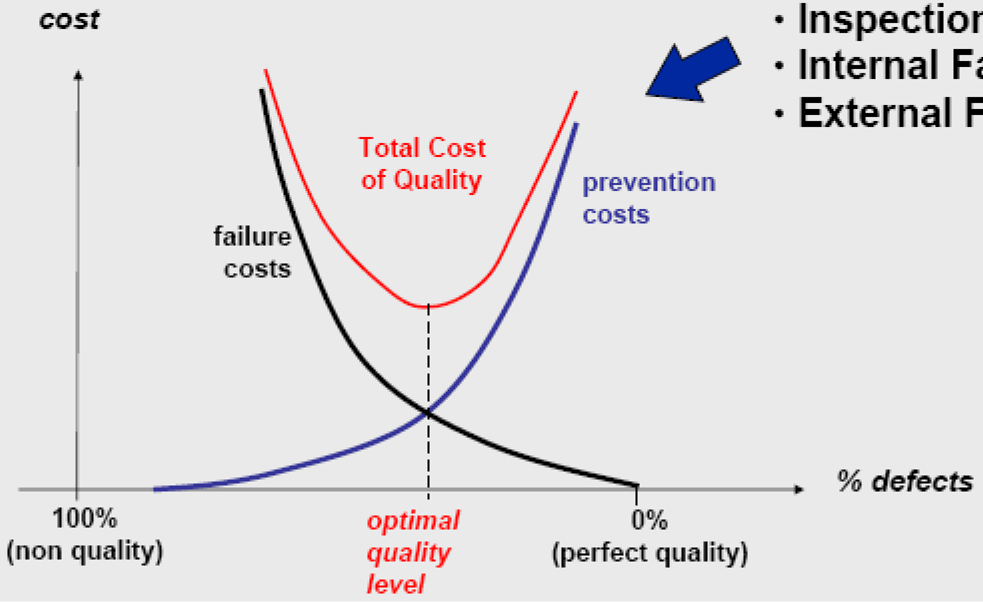
Alternative analysis

- Identify and evaluate options in order to select the cost effective approaches.
- Cost trade-offs and risks should be considered, such as make versus buy, buy versus lease, and the sharing of resources in order to achieve optimal costs for the project. Ex: hire someone else to do the work, because the needed skill set doesn't exist within the organization.

Cost of Quality (CoQ)

- Evaluating the cost impact of quality.
  - Cost of Conformance: Cost to ensure quality (e.g., training, prevention, inspection)
  - Cost of Nonconformance: Cost when things go wrong (e.g., defects, failures, penalties)
- Consider short-term cost reductions versus the implication of more frequent problems later

Cost of Quality



- Prevention
- Inspection
- Internal Failure
- External Failure

Tools & Techniques: Data Analysis

Contingency Reserve

- Reserves are added to costing to cover **identified risks,** cost overruns an error associated with costing.
- The contingency reserve may be a **percentage** of the estimated cost, a **fixed number**, or may be developed by using quantitative analysis methods.
- As more precise information about the project becomes available, the contingency reserve may be **used, reduced, or eliminated.**

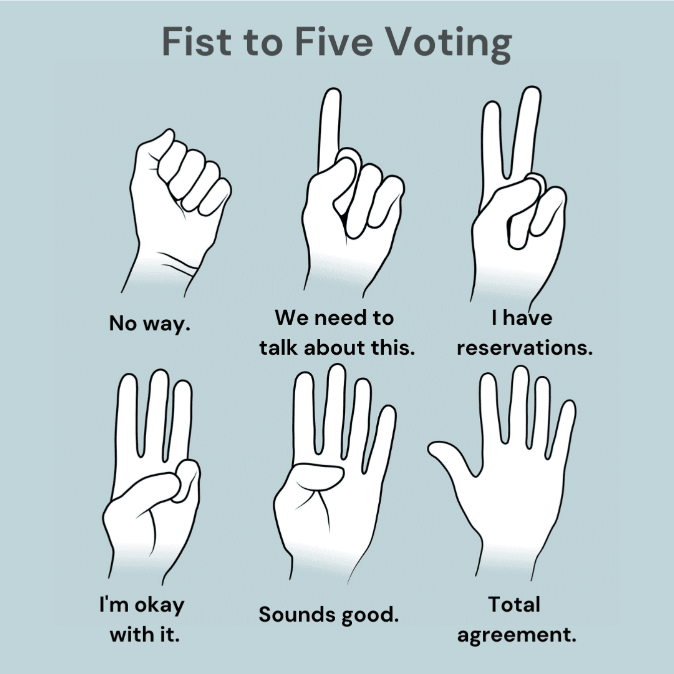
Tools & Techniques

Expert judgment will come from your project team members who are familiar with the work that has to be done. If you don't get their opinion, then there's a huge risk that your estimates will be wrong!

Decision-making techniques help the team decide on the best estimates for the activities they've defined.

Meetings help teams to work together when they estimate the work.

- In Agile method, sprint or iteration planning meetings to discuss prioritized product backlog items (user stories)



Outputs

Cost Estimates

- A cost estimate is a quantitative assessment of the likely costs for resources required to complete the activity.
- Cost estimates may be presented at the activity level or in summary form.
- It is a prediction that is based on the information known at a given point in time.
- Cost estimates should be reviewed and refined during the course of the project to reflect additional detail as it becomes available and assumptions are tested.

ACTIVITY COST ESTIMATES

Project Title: ______________________ **Date Prepared:** ______________________

WBS ID	Resource	Labor Costs	Physical Costs	Reserve	Estimate	Method	Assumptions/ Constraints	Basis of Estimates	Range	Confidence Level

Cost Estimates Worksheet

COST ESTIMATING WORKSHEET

Project Title: ______________________ Date Prepared: ______________________

Parametric Estimates				
ID	Cost Variable	Cost per Unit	Number of Units	Cost Estimate

Analogous Estimates					
ID	Previous Activity	Previous Cost	Current Activity	Multiplier	Cost Estimate

Three-Point Estimates					
ID	Optimistic Cost	Most Likely Cost	Pessimistic Cost	Weighting Equation	Expected Cost Estimate

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BOTTOM-UP COST ESTIMATING WORKSHEET

Project Title: ______________________ Date Prepared: ______________________

ID	Labor Hours	Labor Rate	Total Labor	Material	Supplies	Equipment	Travel	Other Direct Costs	Indirect Costs	Reserve	Estimate

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Outputs

Basis of estimates

Supporting detail must be organized and documented to show how the estimates were created:

- Document basis of estimate (how it was developed)
- Information on the assumptions and constraints
- Information on the range of variance in the estimate
- Indication of the confidence level of the final estimate

Project Document Updates

- **Assumption log.** new assumptions, new constraints, and existing assumptions or constraints may be revisited and changed
- **Lessons learned register. new** cost estimates techniques
- **Risk register.** appropriate risk responses

Determine Budget

Determine Budget

- **What**: process of aggregating the estimated costs of individual activities or work packages to establish an *authorized cost baseline*.
  - Project budget includes all the funds authorized to execute the project.
  - The cost baseline is the approved version of the time-phased project budget that includes contingency reserves, but ***excludes management reserves***.
- **Why:** a realistic and acceptable cost baseline will conforms to financial condition of project.
- **When:** performed once or at predefined points in the project.

Process

Inputs

.1 Project management plan
- Cost management plan
- Resource management plan
- Scope baseline

.2 Project documents
- Basis of estimates
- Cost estimates
- Project schedule
- Risk register

.3 Business documents
- Business case
- Benefits management plan

.4 Agreements

.5 Enterprise environmental factors

.6 Organizational process assets

Tools & Techniques

.1 Expert judgment

.2 Cost aggregation

.3 Data analysis
- Reserve analysis

.4 Historical information review

.5 Funding limit reconciliation

.6 Financing

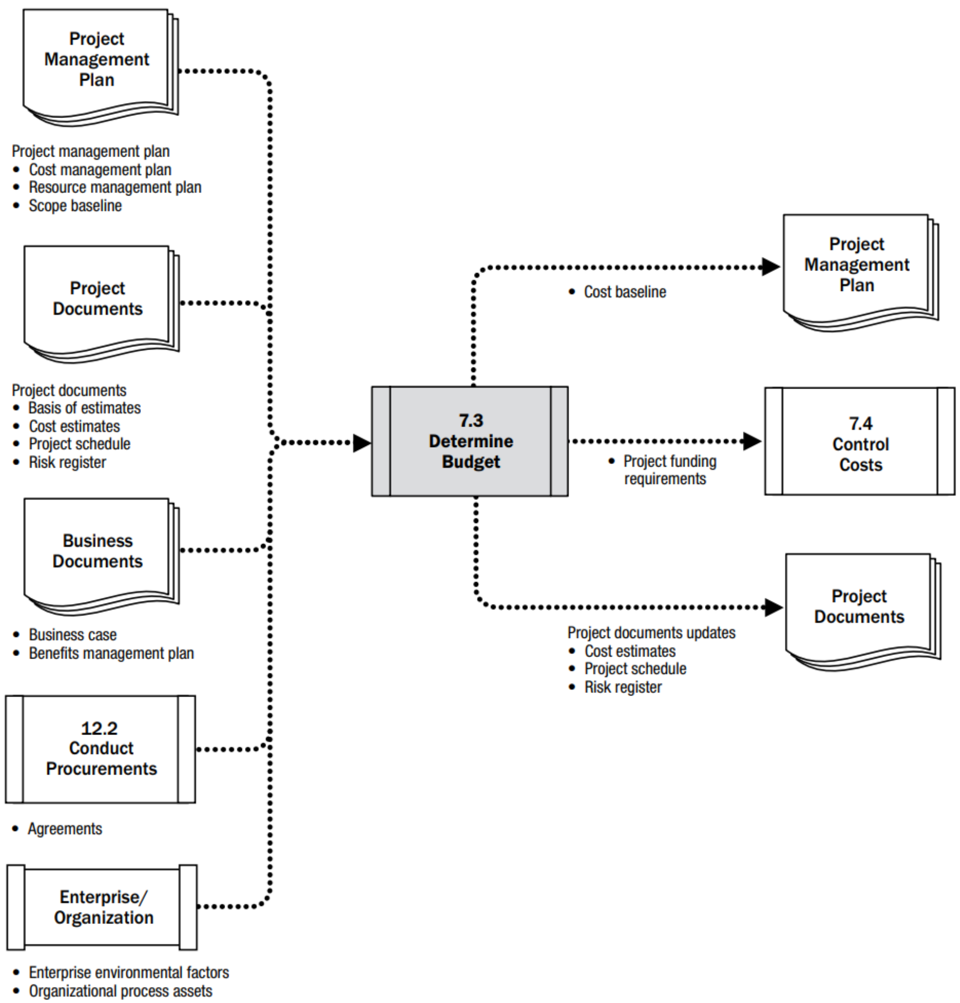
Outputs

.1 Cost baseline

.2 Project funding requirements

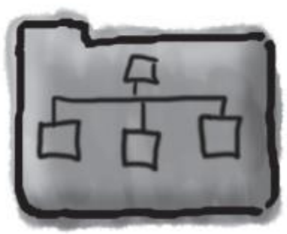
.3 Project documents updates
- Cost estimates
- Project schedule
- Risk register

Data Flow


Project Management Plan
Project management plan
• Cost management plan
• Resource management plan
• Scope baseline
Project Documents
Project documents
• Basis of estimates
• Cost estimates
• Project schedule
• Risk register
Business Documents
• Business case
• Benefits management plan
12.2 Conduct Procurements
• Agreements
Enterprise/ Organization
• Enterprise environmental factors
• Organizational process assets
7.3 Determine Budget
• Cost baseline
Project Management Plan
• Project funding requirements
7.4 Control Costs
Project documents updates
• Cost estimates
• Project schedule
• Risk register
Project Documents

❶ Roll up your estimates into control accounts.

This tool is called **cost aggregation**. You take your activity estimates and roll them up into control accounts on your work breakdown structure. That makes it easy for you to know what each work package in your project is going to cost.



❷ Come up with your reserves.

When you evaluate the risks to your project, you will set aside some cash reserves to deal with any issues that might come your way. This tool is called **data analysis**.

❸ Use your expert judgment.

Here's where you compare your project to historical data that has been collected on other projects to give your budget some grounding in real-world **historical information**, and you use your own expertise and the expertise of others to come up with a realistic budget to cover your project's costs.

❹ Make sure you haven't blown your limits.

This tool is **funding limit reconciliation**. Since most people work in companies that aren't willing to throw unlimited money at a project, you need to be sure that you can do the project within the amount that your company is willing to spend.

❺ Secure funding for your project.

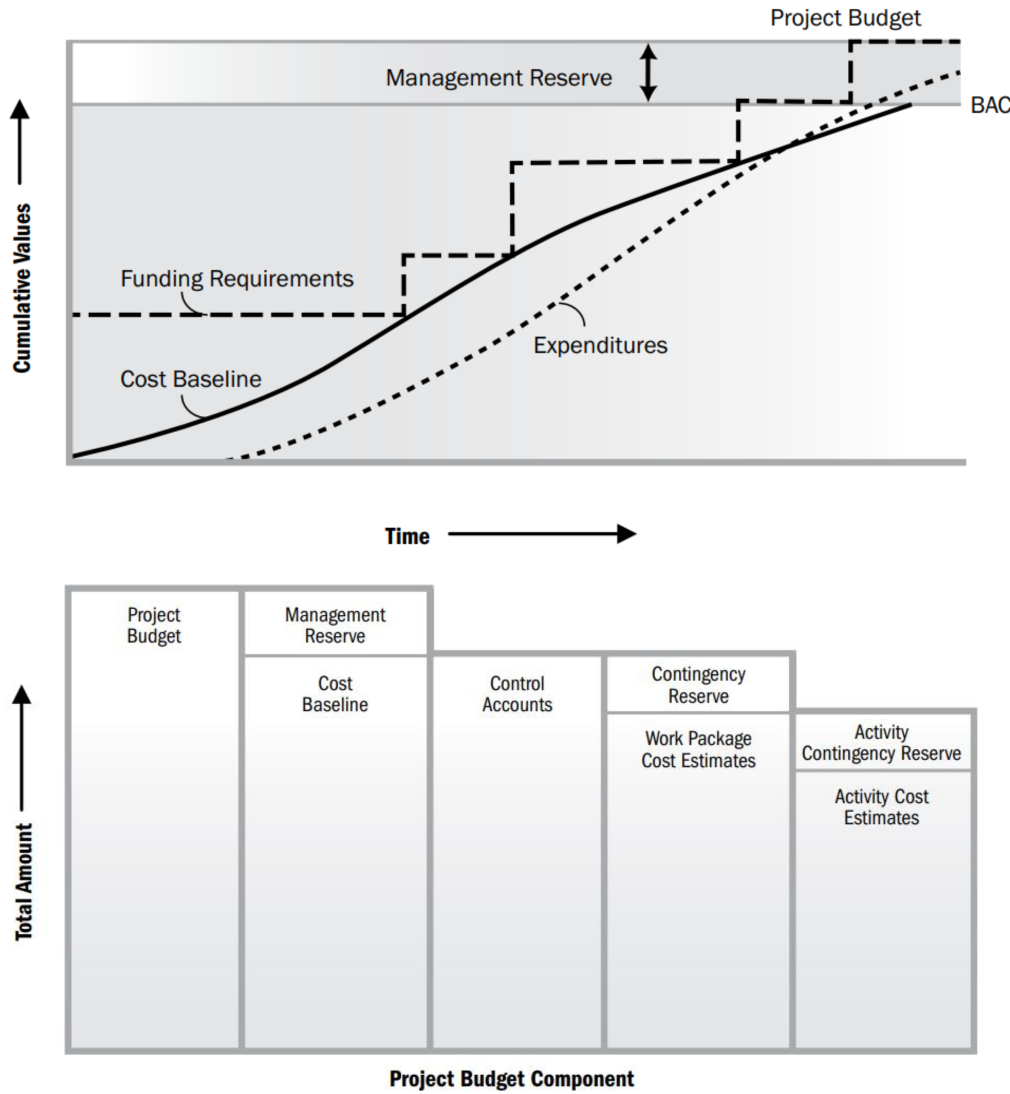
This tool is **financing**. Since some companies go to external organizations to fund specific projects, you might need to meet special external requirements to get the financing. In this tool, you make sure you meet those requirements and get the finance commitments to make your project a success.

Outputs

Cost Baseline

Authorized time-phased budget used to measure, monitor and control overall cost performance of the project.

- Cost baselines forms the shape of an S-curve indicating low spending in the initial stages of project and increasing towards end of the project.
- Management reserves (cost for **unknown-unknowns** such as emergencies are excluded from CB, but included in total budged.



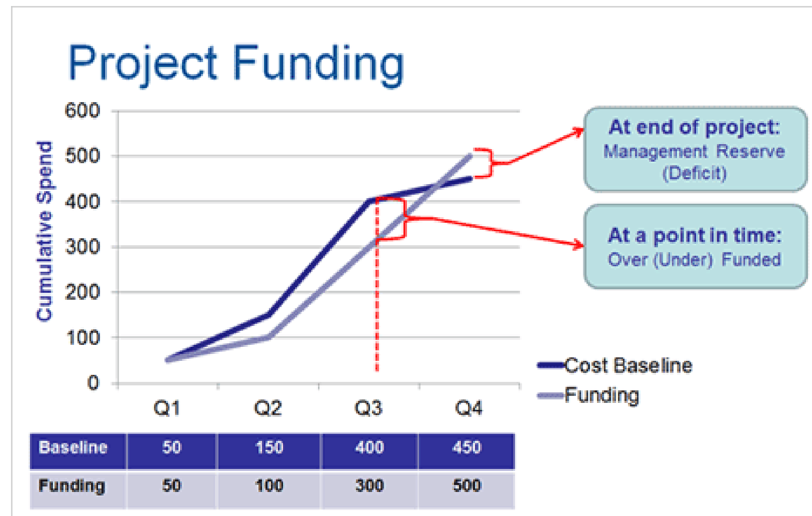
Outputs

Project Funding Requirements

- Funding requirements are derived from cost baselines
- Funding often occurs in incremental rather than continuous
- Total funds required are cost baseline plus management reserve, if any.

Project Document Updates

- Cost estimates
- Project schedule
- Risk register



Control Cost

Control Cost

- **What**: process of monitoring the status of the project to update the project costs and managing changes to the cost baseline.
  - Updating the budget requires knowledge of the actual costs spent to date.
  - Analyzing the relationship between the consumption of project funds and the work being accomplished for such expenditures
- **Why:** maintain cost baseline throughout the project.
- **When:** throughout the project.

Process

Inputs

.1 Project management plan
- Cost management plan
- Cost baseline
- Performance measurement baseline

.2 Project documents
- Lessons learned register

.3 Project funding requirements
.4 Work performance data
.5 Organizational process assets

Tools & Techniques

.1 Expert judgment
.2 Data analysis
- Earned value analysis
- Variance analysis
- Trend analysis
- Reserve analysis

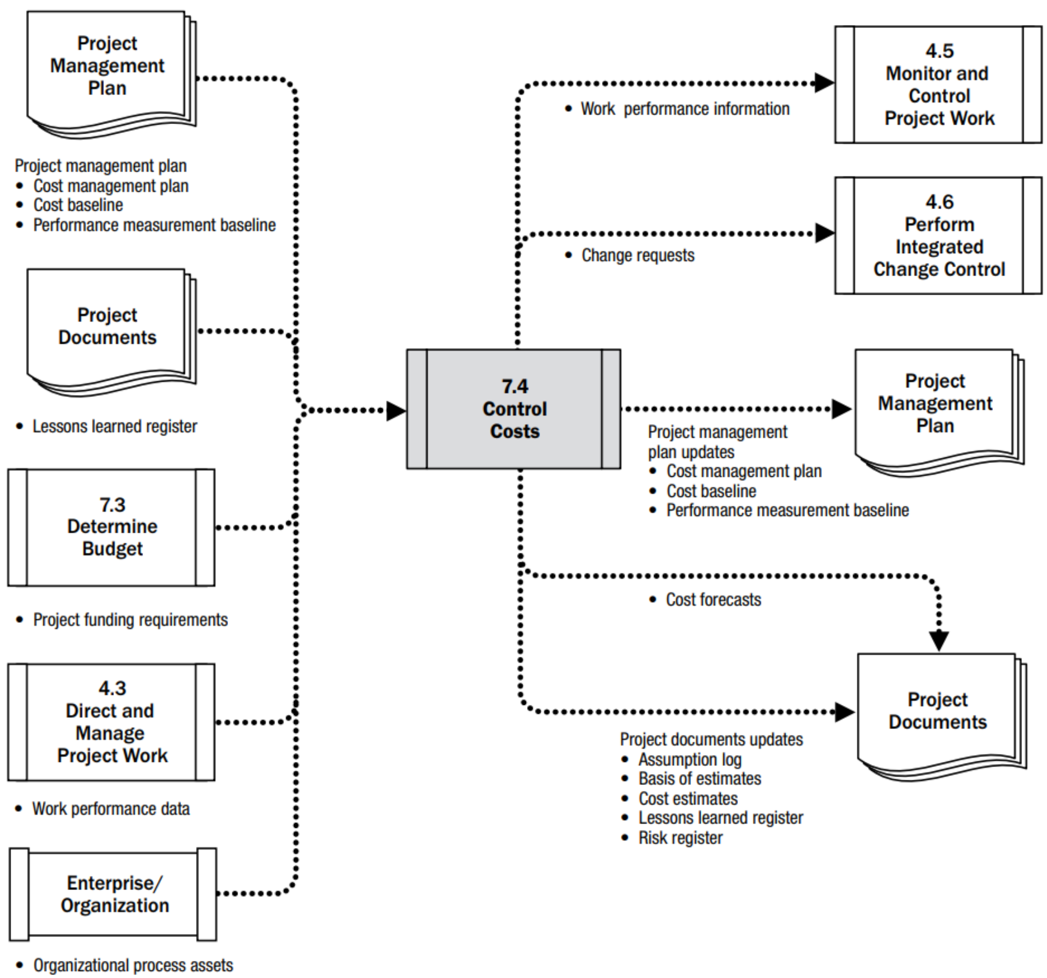
.3 To-complete performance index
.4 Project management information system

Outputs

.1 Work performance information
.2 Cost forecasts
.3 Change requests
.4 Project management plan updates
- Cost management plan
- Cost baseline
- Performance measurement baseline

.5 Project documents updates
- Assumption log
- Basis of estimates
- Cost estimates
- Lessons learned register
- Risk register

Data Flow



Project Management Plan
Project management plan
• Cost management plan
• Cost baseline
• Performance measurement baseline
Project Documents
• Lessons learned register
7.3 Determine Budget
• Project funding requirements
4.3 Direct and Manage Project Work
• Work performance data
Enterprise/ Organization
• Organizational process assets
7.4 Control Costs
4.5 Monitor and Control Project Work
• Work performance information
4.6 Perform Integrated Change Control
• Change requests
Project Management Plan
Project management plan updates
• Cost management plan
• Cost baseline
• Performance measurement baseline
• Cost forecasts
Project Documents
Project documents updates
• Assumption log
• Basis of estimates
• Cost estimates
• Lessons learned register
• Risk register

Tools & Techniques: Data Analysis

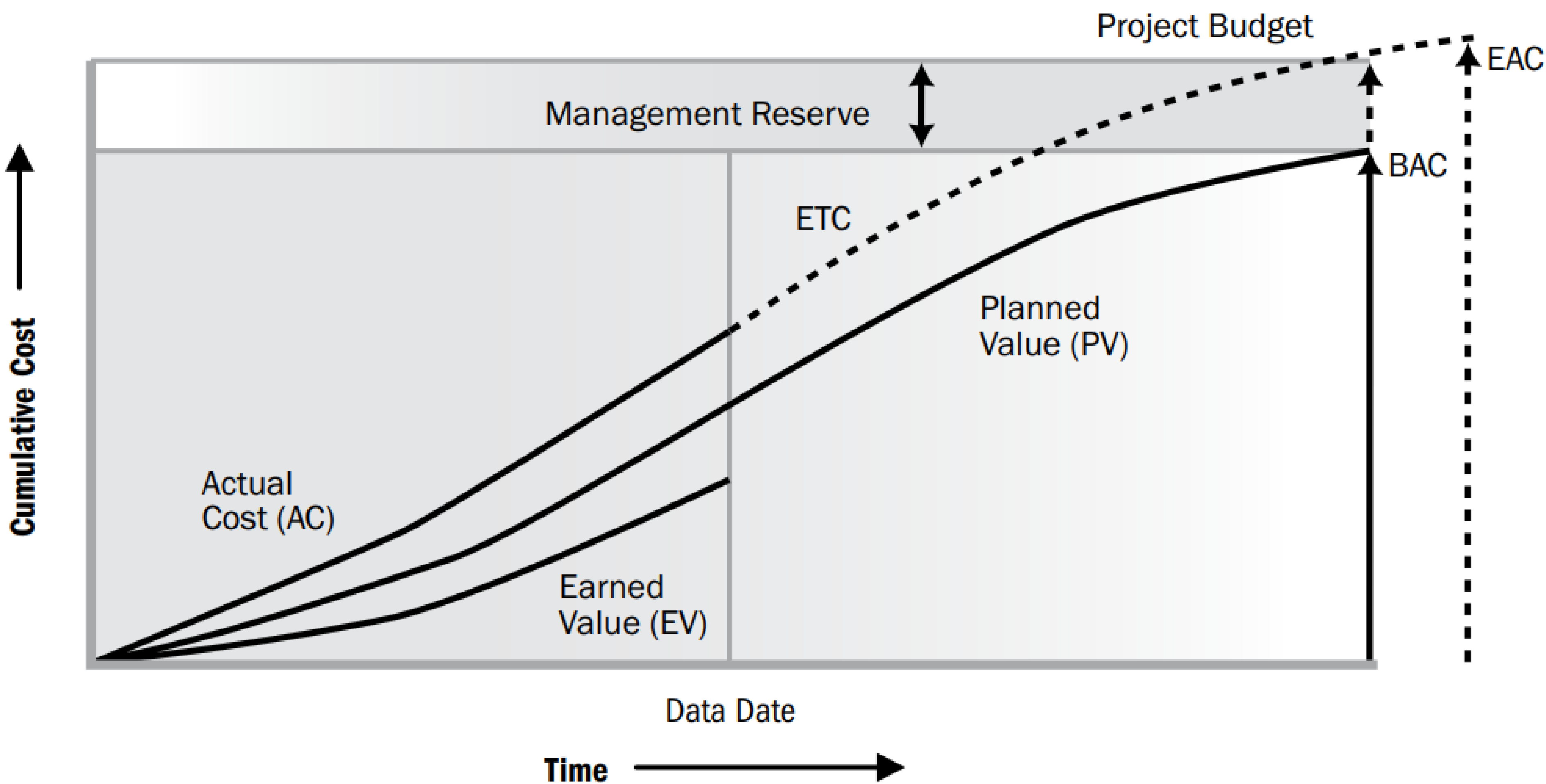
Earned Value Analysis

Compares the performance measurement baseline (scope + cost + schedule) to the actual schedule and cost performance.

- PV, EV and AC can be monitored and reported on both a period-by-period basis (typically weekly or monthly) and on a cumulative basis.
- Accurate picture of project status – Cost, schedule, and technical
- Early identification of trends and problems
- Enables project manager to make informed decisions based on facts
- Results in successful projects: **On time, in budget**

- *Planned value (PV)*: authorized budget assigned to scheduled work.
- *Earn value (EV)*: the budget associated with the authorized work that has been completed.
- *Actual cost (AC)*: total cost incurred in accomplishing the work that the EV measured.

EV Trend



Project Budget
EAC
Management Reserve
BAC
ETC
Planned
Value (PV)
Actual
Cost (AC)
Earned
Value (EV)
Cumulative Cost
Data Date
Time

Planned value

1 *First, write down your*

BAC—Budget at completion

This is the ***first number you think of*** when you work on your project costs. It's the **total budget** that you have for your project—how much you plan to spend on your project.

BAC x

The name "BAC" should make sense—it's the budget of your project when it's complete!

2 *Then multiply that by your*

Planned % complete

If the schedule says that your team should have done 300 hours of work so far, and they will work a total of 1,000 hours on the project, then your planned % complete is 30%.

BAC x Planned % complete

Planned % complete is easy to work out, as it's just the calculation Given amount ÷ Total amount.

3 *The resulting number is your*

PV—Planned value

This is how much of your budget you planned on using so far. If the BAC is $200,000, and the schedule says your planned % complete is 30%, then the planned value is $200,000 × 30% = $60,000.

BAC x Planned % complete = PV

BAC x Planned % complete = PV

PV = BAC x Planned % complete

Earned Value

The schedule says we should have delivered this by now...

Your schedule tells you a lot about where you are supposed to be right now.

...but we only delivered this.

BAC x

BAC x Actual % complete

If your team actually got 35% of the work done when the schedule says they should only have gotten 30% done, that means they're more efficient than you planned!

BAC x Actual % complete = EV

Again, you might see the earned value formula flipped around and written with the EV out front, but remember, it's exactly the same formula.

EV = BAC x Actual % complete

Variance Analysis

Is your project behind or ahead of schedule?

Schedule performance index (SPI)

If you want to know whether you're ahead of or behind schedule, use SPIs. The key to using this is that when you're ***ahead of schedule***, you've ***earned more value*** than planned! So **EV will be bigger than PV**.

To work out your SPI, you just divide your EV by your PV.

$$SPI = \frac{EV}{PV}$$

If SPI is greater than 1, that means EV is bigger than PV, so you're ahead of schedule!

If SPI is less than 1, then you're behind schedule because the amount you've actually worked (EV) is less than what you'd planned (PV).

Schedule variance (SV)

It's easy to see how variance works. The **bigger the difference** between ***what you planned*** and ***what you actually earned***, the **bigger the variance**.

So, if you want to know how much ahead or behind schedule you are, just subtract PV from EV.

Remember, for the sponsor's benefit, we measure this in dollars...

$$SV = EV - PV$$

...so if the variance is positive, it tells you exactly how many dollars you're ahead. If it's negative, it tells you how many dollars you're behind.

Variance Analysis

Are you over budget?

Cost performance index (CPI)

$$CPI = \frac{EV}{AC}$$

If you want to know whether you're over or under budget, use CPI.

Cost variance (CV)

$$CV = EV - AC$$

This tells you the difference between what you planned on spending and what you actually spent.

So, if you want to know how much under or over budget you are, just take AC away from EV.

Remember what CV means to the sponsor: EV says how much of the total value of the project has been earned back so far. If CV is negative, then she's not getting good value for her money.

To-complete performance index (TCPI)

This tells you how well your project will need to perform to stay on budget.

We'll talk about this in just a few pages...

$$TCPI = \frac{(BAC-EV)}{(BAC-AC)}$$

You're within your budget if...

CPI is greater than or equal to 1 and CV is positive. When this happens, your actual costs are less than earned value, which means the project is delivering more value than it costs.

You've blown your budget if...

CPI is less than 1 and CV is negative. When your actual costs are more than earned value, that means that your sponsor is not getting her money's worth of value from the project.

Name	Formula	What it says	Why you use it
BAC—Budget at completion	No formula—it's the project budget	How much money you'll spend on the project	To tell the sponsor the total amount of value that he's getting for the project
PV—Planned value	$PV = BAC \times$ Planned % complete	What your schedule says you should have spent	To figure out what value your plan says you should have delivered so far
EV—Earned value	$EV = BAC \times$ Actual % complete	How much of the project's value you've really earned	To translate how much work the team's finished into a dollar value
AC—Actual cost	What you've actually spent on the project	How much you've actually spent so far	The amount of money you spend doesn't always match the value you get!
SPI—Schedule performance index	$SPI = \frac{EV}{PV}$	Whether you're behind or ahead of schedule	To figure out whether you've delivered the value your schedule said you would
SV—Schedule variance	$SV = EV - PV$	How much behind or ahead of schedule you are	To put a dollar value on exactly how far ahead or behind schedule you are
CPI—Cost performance index	$CPI = \frac{EV}{AC}$	Whether you're within your budget or not	Your sponsor is always most interested in the bottom line!
TCPI—To-complete performance index	$TCPI = \frac{BAC-EV}{BAC-AC}$	How well your project must perform to stay on budget	To forecast whether or not you can stick to your budget
CV—Cost variance	$CV = EV - AC$	How much above or below your budget you are	Your sponsor needs to know how much it costs to get him the value you deliver

Interpret CPI and SPI

- If CPI and SPI are both ***very close to 1***, that means you're delivering the value you promised.
  - A lot of PMOs have a rule where a CPI or SPI between 0.95 and 1.10 is absolutely fine!
- A project that's ***ahead of schedule*** or ***under budget*** will have lower numbers.
- A project that's ***behind schedule*** or ***over budget*** will have lower numbers.

Ahead of schedule
or under budget

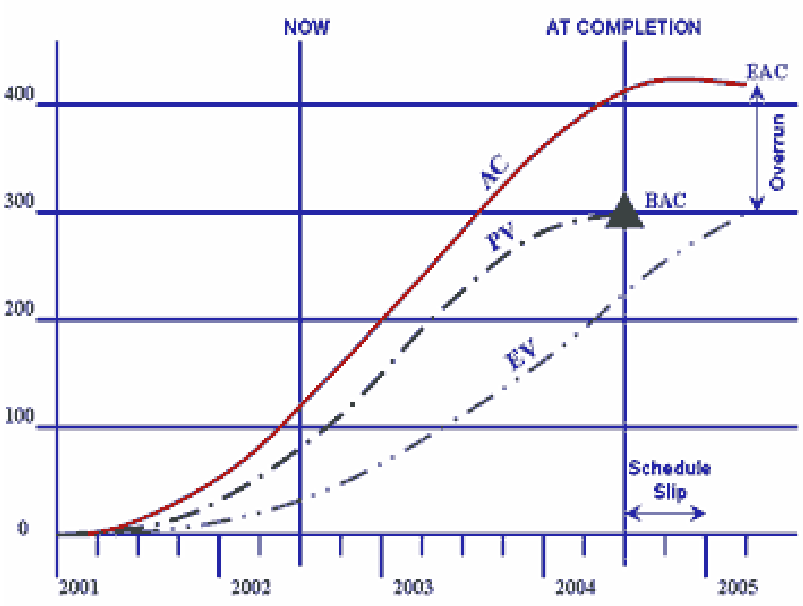
Behind schedule
or over budget

Lower = Loser

Tools & Techniques: Data Analysis

Trend analysis and Forecast

- **Estimate at completion (EAC)**
  - How much do we need at completion?
- **Estimate to complete (ETC)**
  - How much do we need more to complete?
- **Variance At Completion**
  - VAC = BAC – EAC



Trend analysis and Forecast

Scenario 1. The changes project experience will continue to occur for remaining work.

- ETC = EAC – AC
- EAC = BAC/CPI

Current CPI is ***normal*** *(when current variances are thought to be typical)*

Scenario 2. There will be no variation for remaining work and will progress as planned before

- ETC = BAC – EV
- EAC = AC + ETC

Current CPI is ***abnormal*** *(when current variances are thought to be atypical)*

Scenario 3. Original estimate is no longer valid (Original estimate if fundamentally **flawed (floored)).**

- ETC = New estimate for remaining work
- EAC = AC + ETC

Scenario 4. Project is **over budget** but has to **meet a deadline**

- ETC = (BAC-EV)/(CPI x SPI)
- EAC = AC + (BAC-EV)/(CPI x SPI)

Here team considers that remaining work will be completed at the same efficiency rate considering cost and schedule performance

Tools & Techniques: Data Analysis

To Complete Performance Index (TCPI)

Target that your CPI would have to hit in order to hit your forecasted completion cost.

- Based on BAC (within budgeted)
- Re-estimate EAC (over budget)

A high TCPI means a tight budget

- TCPI > 1: more work must be achieved per every dollar spent in the future
- TCPI < 1: lesser work must be achieved per every dollar spent in the future

BAC based:

$$TCPI = \frac{(BAC-EV)}{(BAC-AC)}$$

How much budgeted work is left divided by how much budgeted money is left

EAC based:

$$TCPI = \frac{(BAC-EV)}{(EAC-AC)}$$

How much budgeted work is left divided by how much estimated money is left

Tools & Techniques

Reserve analysis

- As work on the project progresses, these reserves may be used as planned **to cover the cost of risk responses or other contingencies.**
- If the identified risks do not occur, the unused contingency reserves may **be removed from the project budget** to free up resources for other projects or operations.
- Additional risk analysis during the project may reveal a need to request that **additional reserves be added** to the project budget.

Expert Judgment

- Provides valuable insight about the environment and information from previous similar projects.

Project management information system (PMIS)

- Helps to display graphical trends, and to forecast a range of possible final project results.

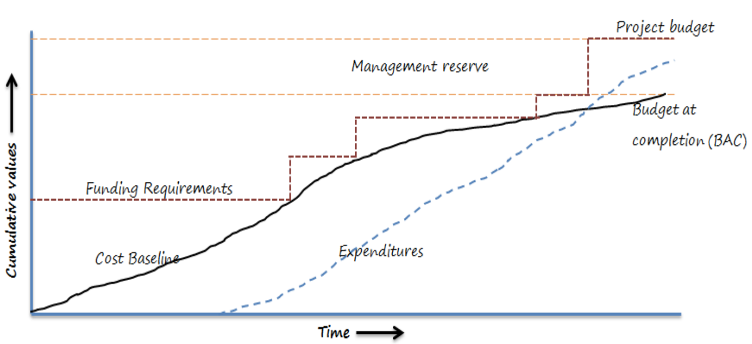
Outputs

Work performance information

- CV, SV, CPI, SPI, TCPI, and VAC values for WBS components

Cost forecasts

- EAC, ETC



Project management plan updates

- **Cost baseline**
- **Cost management plan**
- **Performance Measurement Baseline:**
  - Each of the triple constraints (time, cost, scope) have a baseline, which is established as part of the overall project management plan.
  - These three baselines put together are called the Performance Measurement Baseline.